

CITY BRAIN CONVERGENCE / PHASE 1

CIVIS: 5 October deliverables

Incident responses, selected actuator risks, and data conventions

Triplet 04: CIVIS (Brain), Trinity (Twin), 9antra (Guardian)

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Status: CIVIS submission for partner review. The task and deadline are agreed; this catalogue has not yet been verified by all three teams.

1 Scope and authority

This document completes the written CIVIS work assigned in section 11 of the final decision report: (A) the incident/response catalogue for all-team verification; (B) selection of three actuator types for risk levels; and (C) CIVIS data conventions. It documents behavior and formats; it does not claim implementation or successful integration.

Deliverable	Content	Acceptance owner
A: incidents	Eleven supported incident types, evidence, responses, and limits	CIVIS writes; all verify
B: actuator risks	Dispatch, water-valve control, power-grid switching	CIVIS selection; partners implement
C: data conventions	Time, units, locations, IDs, missing data, examples, validation	CIVIS issues; partners check compatibility

Domain scope. Traffic, Water, Power, Air Quality, and Emergency. A shared node may belong to several domains. Land supplies map/topology context; it is not a sensor domain. Telecom and Waste are outside this deliverable. The final report still schedules them for v1.0 on 24 October; excluding them now does not silently change that later agreement.

Source precedence. The user's current five-domain scope governs this package. The final decision report, updated 2 October, governs triplet choices. The official Phase 1 specifications govern challenge requirements. Phase 0 and the first-session summary supply rationale only where later choices do not supersede them. Conflicts are listed explicitly in section 6.

Not due from CIVIS on 5 October. Dali owns the SAC email. Trinity proposes the domain caps and 9antra confirms them. Retry semantics are open for 9 October; the failure table and Test Book have later dates; the Interface Card is proposed for 24 October. No numerical caps, score cut-offs, TTLs, retry waits, or timeouts are invented here.

2 Rules inherited from the final decisions

The following constrain every incident response in this package (final report, sections 2–7).

1. The Brain detects incidents and requests responses. Only the Twin changes the simulated physical city. The Brain does not bypass the generic `actuate` tool.
2. Every physical action, including an untiered traffic-signal change, needs Guardian approval for that exact action. The Twin enforces the token check. Cached or pushed trust scores are observations, not authorization.
3. Guardian scores are numeric only. Trusted, Degraded, and Untrusted are derived using Guardian-published cut-offs; Isolated is a containment state, not a score band. A numeric score never turns an isolated device back into a usable one.
4. Only Guardian requests `isolate_sensor`, `quarantine_device`, or `rollback_reading`. Brain may report suspicious evidence and withhold its own action, but cannot invoke containment.
5. A repeated `idempotency_key` returns the original result rather than re-executing. Distinct actions in a plan use distinct keys; transport retransmission of one unchanged action reuses its key.
6. The Twin applies domain caps regardless of trust score. The Power exception concerns isolation of all Power sensors; it does not authorize opening every grid switch.
7. High-risk actions receive a dry-run. Two-step confirmation is a working default: a pending response is not completion. The unresolved pending/retry timing details remain open.
8. A corrected reading uses the last trusted *reported* value, not the Twin's hidden true state. The Brain never reads simulator ground truth to make an operational decision.
9. Mock tokens may be unsigned if signing is not ready. Such a demonstration must be labelled non-tamper-proof. Ed25519 is agreed for v1.0 on 24 October.

2.1 Incident decision sequence

Validate and normalize evidence; exclude unavailable, isolated, or rejected evidence; detect and correlate the incident; select feasible responses; obtain action-specific approval; ask the Twin to execute; and record the result. For a high-risk response, obtain a satisfactory preview before requesting final approval. If the preview changes action parameters, the token must approve those revised parameters.

Every catalogue entry assumes this sequence. A real incident can remain active while an unsafe or unsupported response is withheld. **No action is sometimes the correct response**; it must have an explicit reason, not disappear from the timeline.

2.2 Detection parameters

Use named, configurable per-device or per-node bounds and persistence windows. The following notation specifies dependencies without pretending the numbers were agreed:

- `traffic_high_pct`, `water_high_cm`, `water_low_cm`;
- `power_high_kw`, `power_low_kw`, `pm25_high_ug_m3`;
- `detection_window_ticks`, `freshness_limit_ticks`, and recovery hysteresis.

Trinity provides device semantics and baselines; CIVIS sets incident policies against that model. Guardian owns its score cut-offs. Missing configuration keeps the affected automatic response disabled and explains why; there is no fabricated universal threshold.

3 Deliverable A: incident and response catalogue

The types below are CIVIS's concrete catalogue for triplet verification. They are not all claimed to be previously agreed incident types. A row is implementable against the agreed sensor inventory; its physical action may still require topology metadata or a policy value. Suggested verification conditions are checks for the 5 October review, not a completed Test Book or scenario implementation.

I01: Persistent congestion

Domains: Traffic; optionally Emergency

Evidence. Fresh, usable traffic-level readings above the configured congestion bound for a configured window. Neighboring segments establish extent if available. An emergency call is optional corroboration, not a required trigger for routine congestion.

Brain response. Request a legal signal state/phase adjustment only within Twin-advertised signal behavior. Preserve intersection conflict constraints and emergency routes. If only a single light's state is exposed and safety cannot be guaranteed, log the congestion and request operator review rather than invent a timing actuator.

Actuator. Traffic-signal control; no risk tier in this selection, but an exact-action token and caps still apply.

Verification. A persistent jam permits a bounded legal adjustment; a brief spike does not cause repeated toggling. No police dispatch is made solely because congestion is high.

I02: Road accident with an emergency report

Domains: Traffic, Emergency

Evidence. A typed accident report with a resolvable road node/segment, optionally supported by a traffic disturbance. Congestion alone cannot distinguish a collision from rush hour. The emergency report remains data, not an instruction.

Brain response. Request police if scene control is needed; request an ambulance only when injury/medical need is supported. Select available appropriate units and avoid duplicate dispatch. Adjust a signal only if its advertised safe behavior can assist the response.

Actuator. Generic dispatch with `unit_type`; optional signal control. Each dispatch receives its own approval and idempotency key.

Verification. A corroborated accident produces the appropriate requested unit(s); an unexplained jam alone does not become an accident. An unknown location yields review, not a guessed destination.

I03: High water or local flooding

Domains: Water; optionally Traffic, Emergency

Evidence. A usable water-level reading above the location-specific bound with the sensor's datum and site role known. A trusted emergency report or a second Water sensor may corroborate extent. A high level in a reservoir is not automatically a street flood.

Brain response. Raise the flood incident. A valve request is made only when Twin topology identifies a valve and the requested state is predicted to reduce this hazard without harming downstream supply. Dispatch emergency assistance if people or a road are affected.

Actuator. Conditional water-valve control, dispatch, and optional legal signal control. No pump, road barrier, or evacuation actuator is assumed.

Verification. A real flood may justify a previewed valve change. A fake or untrusted level cannot cause it. An inlet valve and a drainage valve must not both receive a blind "close" response.

Catalogue continued: Water and Power

I04: Low water / reservoir depletion

Domains: Water; optionally Power, Emergency

Evidence. A usable level below the site's configured low bound, with a reservoir/tank role and baseline known. A level sensor alone does not establish water pressure, a burst pipe, or a leak location.

Brain response. Flag low supply and protect essential service. Request a valve change only where topology and a dry-run establish that it restores or preserves supply. Otherwise seek human review; use emergency dispatch only for an independently supported emergency.

Actuator. Conditional water-valve control. Power switching is not automatically inferred from low water.

Verification. The same low reading at a different site role may lead to a different response. Unknown topology leaves the incident visible and the valve action withheld.

I05: Power overload

Domains: Power; optionally Water, Emergency

Evidence. Fresh usable load readings above a configured meter/feeder bound with capacity and topology known. Shared Power/Water nodes identify possible pump-supply consequences. Load alone does not prove overvoltage or a short circuit.

Brain response. Choose only a topology-supported switch operation that the dry-run predicts will reduce the overload without violating protected-service constraints. If there is no safe candidate, withhold switching and escalate. Dispatch emergency units only if an actual hazard is also reported.

Actuator. Power-grid switching; dispatch if separately justified. The Brain cannot invoke the Power sensor-isolation exception.

Verification. A genuine overload can produce a bounded, approved switch request. An untrusted load spike cannot. A candidate that removes an essential pump or emergency service is rejected or needs an explicit agreed exception.

I06: Possible loss of electrical service

Domains: Power; optionally Water, Emergency

Evidence. A sustained load drop compared with an expected baseline, corroborated by a usable service-loss report or modeled downstream effect. Zero load alone is not proof of an outage; it may mean zero demand, a failed meter, or a deliberately open switch.

Brain response. Maintain an outage hypothesis until corroborated. Request restoration only when switch state, topology, and preview show a safe, available path. Withhold blind reclosing. If a critical service is affected, show a human-review escalation and use dispatch only for the resulting human emergency.

Actuator. Conditional power-grid switching. Commanded/actual-state feedback is not assumed available in the early mock.

Verification. A dead meter must not produce an automatic reclose. Where actual-state feedback is absent, record "requested" or "unverified", not "power restored".

Catalogue continued: Air Quality and Emergency

I07: Air-quality deterioration

Domains: Air Quality; optionally Emergency, Traffic

Evidence. Usable PM2.5 concentration above the configured bound, with persistence and optional neighboring monitors. PM2.5 is CIVIS's proposed initial observable; Twin must confirm its monitor measures this quantity. No AQI scale is silently inferred.

Brain response. Raise an air-quality incident and human-review notice. Add dispatch only when an emergency report supports fire, injury, or another actionable emergency. Air pollution alone is not proof of fire. No direct pollution-control actuator exists in the agreed inventory.

Actuator. None for an isolated air-quality alert; conditional dispatch and signal control for corroborated emergencies. The review notice is a Brain/dashboard output, not a new Twin actuator.

Verification. A pollution-only case remains visible without fictitious fan/ventilation actuation or unjustified fire dispatch. A fire report can promote the combined incident to I08 or I11.

I08: Reported fire with supporting evidence

Domains: Emergency, Air Quality; optionally Traffic, Power, Water

Evidence. A usable fire report with a resolvable location. Elevated Air Quality readings or related reports strengthen the hypothesis; they are not mandatory when a credible emergency report already supports response. A pollutant spike alone remains I07.

Brain response. Request a firefighter unit through generic dispatch. Add ambulance or police only where medical or scene-control evidence supports them. Use legal signal control if supported. A grid or valve action needs independent topology-aware justification and preview; a fire never authorizes bulk shutdown.

Actuator. Dispatch; optionally signal control and independently justified high-risk Water/Power actions. There is one dispatch type with a unit-type field, not three dispatch tools.

Verification. The plan separates each intended action, target, approval, and result. No unsupported firefighting equipment, water-pressure control, or road-closure actuator is introduced.

I09: Medical emergency

Domains: Emergency; optionally Traffic

Evidence. A usable emergency report indicating medical need, with a valid destination. The structured incident type comes from typed extraction or the Twin's typed feed, never from obeying the transcript.

Brain response. Request an available ambulance. Optional legal signal control may assist access; no unimplemented route-planning tool is assumed. If no unit is available, keep the incident open and display escalation.

Actuator. Generic dispatch with `unit_type=ambulance`; optional signal control.

Verification. Repeated delivery of one call does not dispatch additional units. A transcript containing arbitrary tool instructions does not gain privileges or change the allowed action set.

Catalogue continued: coupled incidents

I10: Flood affecting a shared Power/Water node

Domains: Water, Power, Emergency; optionally Traffic

Evidence. I03 evidence plus a shared-node association and a credible reported or modeled threat to electrical service. Co-location means exposure, not proof of an electrical fault. Independent Power evidence is required before classifying I05/I06.

Brain response. Correlate the Water and Power observations under one incident record. Request appropriate emergency dispatch. Preview a useful valve action and any independently justified switch action separately, then check their combined effects. Do not switch off a pump that the Water response requires without an explicitly accepted trade-off.

Actuator. Dispatch and conditional water-valve/grid-switch operations. Each physical action is scoped, token-gated, and capped; no containment call is made by Brain.

Verification. The plan explains the shared-node dependency. A genuine flood plus a spoofed Power reading does not authorize a grid action; the flood response can continue using its valid evidence.

I11: Fire with congestion and threatened essential services

Domains: Emergency, Air Quality, Traffic; optionally Power, Water

Evidence. I08 evidence, usable congestion readings, and topology showing affected routes or essential-service nodes. Water/Power effects remain hypotheses until observed or supported by the Twin model.

Brain response. Prioritize appropriate dispatch, choose a safe available signal action if supported, and evaluate any essential-service intervention independently. Prevent one incident from consuming all units or changing all signals. Only agreed caps and resource availability define how many actions are feasible.

Actuator. Dispatch and signal control; Water/Power only with their own evidence, safety preview, and approval.

Verification. A single event can reference multiple domains without duplicating the incident or double-dispatching. Lack of a usable Air Quality monitor does not silently convert all emergency-call evidence into untrusted data.

3.1 Cross-domain coverage check

Domain	Catalogue coverage
Traffic	I01, I02; secondary effects in I03, I08, I09, I11
Water	I03, I04; shared-node dependency in I10, optional I11
Power	I05, I06; shared-node dependency in I10, optional I11
Air Quality	I07, I08, I11
Emergency	I02, I08, I09, I10, I11; conditional in I03–I07

An incident has one `incident_id` and a list of affected domains. Separate calls about one event may be correlated, but deduplication must not merge distinct emergencies solely because they share a node. The correlation policy is an implementation choice, not a numerical deadline commitment.

3.2 Negative cases that verify the responses

These are verification variants applied to the catalogue, not additional real-world incident types. Detection of spoofing, replay, or a stuck sensor remains Guardian's responsibility.

Variant	Expected CIVIS behavior	Important boundary
Healthy baseline	No physical action without an incident or explicit operational reason	"No action" is observable
Untrusted high reading	Exclude it from action evidence; seek alternative usable evidence; escalate when exhausted	Do not isolate the device from Brain
Stuck/replayed/old reading	Do not treat repeated packets as new corroboration	Guardian checks integrity; Brain checks format/freshness
Real event plus bad sensor	Continue justified response using available usable evidence	A bad Power reading need not suppress a credible emergency call
Fake all-clear during a real event	Do not close the incident on rejected or stale recovery evidence	Clear state requires usable recovery evidence
Duplicate action request	Reuse the original key and recognize the original result	No extra unit or repeated physical actuation
Unavailable Guardian	No new physical action without a currently valid action-specific token	No Phase 0 fail-open exception
Unknown location/topology	Preserve incident and explain withheld response	Never guess a valve, grid feeder, or destination
Instruction hidden in call text	Extract facts only; reject unauthorized action fields	Text cannot request isolation, override policy, or approve itself
Actuator result unavailable	Track a request without claiming physical success	Feedback may be absent in early mocks

3.3 What to verify together by 5 October

- Trinity confirms that every named sensor quantity, node relation, and actuator parameter is actually representable in the mock inventory.
- 9antra confirms that incident evidence can be associated with its score-only verdicts and action tokens, including multi-domain evidence.
- All teams confirm conditional responses, unsupported actions, and the absence of Telecom/Waste incidents from this package.
- CIVIS records accept/change/reject for each catalogue ID. No blank review field is counted as approval.

4 Deliverable B: the three risk-rated actuator types

CIVIS selection: dispatch, water-valve control, and power-grid switching. This uses the delegated choice of two or three types (final report, sections 2, 4, 5 and 11). The labels retain useful Phase 0 distinctions, but do not import its old score thresholds or token exemptions.

Actuator type	Class	Rationale and required behavior
Dispatch	R2	Allocates scarce police, ambulance, or firefighter capacity. One risk class applies to generic dispatch as a whole; unit type does not create a different class. Require exact-action approval, valid destination, availability, and applicable caps.
Water-valve control	R3	Can worsen flooding or deprive downstream users. Apply the class to both open and close; reversibility of a state command is not proof of low physical impact. Require role/topology-aware preview and exact-action approval.
Power-grid switching	R3	Can interrupt essential services or reconnect an unsafe segment. Apply the class to both open and close. Require topology-aware preview, protected-service checks, and exact-action approval.

4.1 What receives no risk tier

Traffic-signal control is not selected for a tier in this package. It still needs an action-specific token, safe permitted state transitions, and applicable caps. This selection does not grant an ungated signal-control path.

Guardian containment receives no risk level, as agreed. Only Guardian can request it; the Twin enforces caller permissions and domain caps. The exception for isolating all Power sensors is not a dispatch, valve, or switch exception. Air Quality has no direct actuator; an alert therefore has no physical actuator tier.

4.2 High-risk preview and authorization

R3 is the high-risk category for mandatory preview in this CIVIS selection. R2 dispatch is still gated and resource-checked; a dispatch preview may be used if supported, but Phase 0's blanket R2 preview requirement is not asserted as a triplet agreement. Any additional mandatory preview rule needs an explicit decision.

The Twin should expose authoritative risk metadata in its action listing. A caller cannot lower the tier by supplying a different label. Exact request contents must be bound by the token, including targets and parameters. **Unresolved:** score minimum per risk level, token lifetime/clock basis, cap accounting, and canonical token encoding. Their values belong to the pending shared decisions, not this selection.

4.3 Resource contention

Multiple incidents compete for actual available units. Priority is based on supported human/essential-service consequences, not on which sensor sent the largest raw number. Numeric comparisons across percent, centimeters, kW, and particulate concentration are meaningless. Tie-breaking, capacity, and minimum district coverage need explicit policy values before automation; do not promise dispatch-all behavior.

5 Deliverable C: CIVIS data conventions

These are the conventions CIVIS is issuing under the final report's delegated data-format choice. New representations below are **CIVIS choices for partner compatibility review**, not statements that the earlier meetings already fixed each field. Concrete examples are synthetic and contain no real secrets.

5.1 Encoding and validation boundary

Use UTF-8 JSON with lower-case snake_case keys. Numeric measurements are JSON numbers, not quoted numbers. A Boolean is not accepted as an integer. Reject non-finite values such as NaN/Infinity, duplicate object keys, wrong required types, and missing required keys. Output only documented fields. Ignore unknown extra fields on input; ignoring never gives them meaning, authority, or a place in a signed action.

Unknown values of *required known enums* are errors; they are not the same as unknown extra fields. No silent conversion from "80" to 80, no implicit centimeters-to-meters conversion, and no replacement of a missing measurement with zero. An adapter may perform an explicitly registered, lossless unit conversion while preserving the original reading; otherwise reject the incompatible quantity/unit pair.

5.2 Domain names and measurement conventions

Domain code	Quantity	Unit code	Definition
traffic	traffic_level	percent	Proposed normalized congestion index on a segment: 0 is model-defined free flow, 100 is model-defined saturation. This is not vehicle count, speed, or measured occupancy. Trinity confirms the mapping.
water	water_level	cm	Reported height relative to a declared local datum. Metadata identifies tank, river, reservoir, or street site role. Do not compare different datums blindly.
power	power_load	kW	Active-power load at the meter's declared node/feeder. Capacity and expected baseline are metadata/configuration, not guessed from the unit.
air_quality	pm25	ug/m3	PM2.5 mass concentration. A PM2.5 choice must be confirmed against Twin's monitor; never equate it with an unspecified AQI index.
emergency	emergency_call	None	A typed event object; not a scalar measurement. Store reported location/type and untrusted transcript separately.

Traffic percent has the representational bound $0 \leq v \leq 100$; this is not an incident threshold. PM2.5 concentration is non-negative. Water bounds follow site/datum metadata; Power sign convention follows the declared meter role. Negative readings are not globally clipped, since clipping can hide faults and valid reference levels.

5.3 Clock and event identity

Field or concept	CIVIS rule
timestamp	Simulated observation time as UTC RFC 3339, emitted as YYYY-MM-DDTHH:MM:SS.sssZ. Accept an explicit equivalent offset and normalize to UTC; reject timezone-free strings.
tick	Non-negative integer logical simulation tick. Tick duration is declared in run metadata as positive <code>tick_duration_s</code> ; one tick is not silently assumed to equal one second.
run_id	Non-empty stable string per simulation execution; a seeded replay gets a new run ID, while the seed and model version identify replay conditions.
reading_id	Unique reported observation identity within the run; duplicates with identical identity/content are duplicate delivery. A changed payload under one identity is an integrity conflict, never a fresh observation.
device_id, node_id	Case-sensitive stable identifiers from Twin discovery. Device identity differs from location. An actuator ID must not be substituted for a sensor ID.
seq	Non-negative per-device sequence number, increasing for new emissions in a run. A transport retransmission preserves its sequence/reading identity. Gaps are visible; no silent interpolation.
received_at	Optional wall-clock UTC arrival time, separate from simulated observation time. Used for operational latency, never silently compared with simulated time.

Freshness compares times within the same run and clock domain. Ordering relies on `tick`, then device sequence; timestamps do not provide a total order across producers. Token expiry must use an explicitly agreed clock basis, still open. Do not reuse Phase 0's five-second TTL as if it had been adopted.

5.4 Locations, shared nodes, and metadata

Use `node_id` as the authoritative routing/topology reference. Optional WGS84 location uses named `latitude_deg` and `longitude_deg` fields, latitude in $[-90, 90]$ and longitude in $[-180, 180]$. Avoid ambiguous coordinate arrays; local map coordinates, if used, need a declared coordinate system and units.

`domains` is a non-empty list of unique domain codes, allowing shared nodes such as `["water", "power"]`. The measured `quantity` identifies what this reading observes; a shared node does not make one sensor measure two quantities. Neighbor IDs, device role, water datum, feeder capacity, valve purpose/connectivity, and available dispatch unit types belong to discoverable Twin metadata. Without them, topology-dependent actions remain conditional.

Run metadata must declare the seed, simulation start time, positive tick duration, and model/configuration version. A seed alone does not guarantee reproducibility across different model versions, event orderings, or configurations. Reading identity and device sequence are scoped to that declared run.

5.5 Missing, corrected, and untrusted are different

Representation	Meaning and handling
<code>value:null, quality:"missing"</code>	Sensor emitted a no-value observation. Quantity, unit, IDs, tick, and observation time remain present. It is not zero, normal, or evidence that a hazard ended.
<code>quality:"reported"</code>	Ordinary reported measurement; it says nothing about trust. Guardian may still reject it.
<code>quality:"corrected"</code>	New correction record referencing the affected reading and the last trusted reported source. Preserve the raw record and original source time; a fresh correction time does not make the replacement sample fresh.
No packet at all	Absence after an expected interval; do not fabricate a sensor reading. Expected cadence and timeout policies are separate configuration.
Untrusted or isolated	A trust/containment outcome, not missing-data encoding. Exclude from usable action evidence under Guardian policy; retain records for audit.

Required fields cannot be replaced with null unless their convention explicitly allows it. An optional unknown location may be null, but a dispatch destination cannot be null at actuation time. NaN, empty strings, -999, and the text "unknown" are not missing-measurement encodings.

5.6 Numeric sensor reading: synthetic example

```
{
  "run_id": "run-demo-001",
  "reading_id": "reading-water-17",
  "device_id": "sensor-water-01",
  "node_id": "node-pump-01",
  "domains": ["water", "power"],
  "seq": 17,
  "tick": 120,
  "timestamp": "2026-10-05T08:00:00.000Z",
  "quantity": "water_level",
  "value": 180.0,
  "unit": "cm",
  "quality": "reported"
}
```

This is a reported value, not the Twin's truth and not an approved flood. The number 180 is an example, not the triplet's flood threshold. A missing sample keeps the same shape but has `value:null` and `quality:"missing"`; it needs its own reading identity.

5.7 Emergency event and safe text handling

```
{
  "run_id": "run-demo-001",
  "reading_id": "call-004",
  "device_id": "feed-emergency-01",
  "seq": 4,
  "tick": 120,
  "timestamp": "2026-10-05T08:00:00.000Z",
  "domains": ["emergency"],
  "event": {
    "call_id": "call-004",
    "reported_type": "fire",
    "reported_node_id": "node-pump-01",
    "transcript": "Smoke reported near the pump station."
  }
}
```

Allowed initial `reported_type` values: `fire`, `medical`, `accident`, `flood`, `other`, and `unknown`. These are reported hypotheses, not confirmed incident classes. Unresolved location is `reported_node_id:null`; it triggers clarification/review, never a guessed dispatch.

If Twin only provides text, CIVIS's typed extraction produces this normalized event view and keeps the source text/reading link. The transcript is untrusted content. It cannot set risk levels, targets, authentication, tokens, isolation policy, or tool names. Action selection uses an allow-list and deterministic parameter checks; embedded instructions are not commands.

5.8 Trust score convention and provenance

The proposed canonical score is a finite JSON number in $[0, 1]$, higher meaning more trusted. **9antra must confirm the scale and direction against its existing demo.** If it uses another declared scale, a jointly documented adapter normalizes it; an undeclared scale is rejected. Do not infer a percentage by multiplying or dividing arbitrary values.

The verdict content is a score only. Correlation metadata is separate: it identifies the run, the exact reading(s) or action being assessed, and verdict observation time. Do not add a label, reasons array, or fabricated cut-offs to Guardian's agreed verdict payload. A scalar without an associated subject cannot safely be used for actuation.

For example, the verdict content may be the JSON scalar `0.93`, associated with `reading-water-17` by the request/stream envelope. This does not mean "Trusted" until Guardian's published policy is applied, and it is not an action token.

Brain's internal incident/intent records retain `incident_id`, source `reading_id` references, proposed action, targets, reasoning, and request/result IDs. This is minimum decision provenance, not a new shared log format: Trinity's log draft is due for joint review on 9 October. A reading-level score and an action-approval token have distinct purposes.

5.9 Action parameters under the agreed four fields

The final report fixes `actuate(action, targets, token, idempotency_key)`, but not the shape of `action`. A string-only action name cannot also carry signal state, valve state, or dispatch destination without another agreed mechanism.

CIVIS proposed representation for Trinity confirmation: make `action` an object with `name` and `parameters`, preserving the four outer fields. The action listing supplies the authoritative names, permitted parameters, targets, and risk metadata. Names below are examples for alignment, not claims of implemented MCP tools.

```
{
  "action": {
    "name": "dispatch",
    "parameters": {
      "unit_type": "firefighter",
      "destination_node_id": "node-pump-01"
    }
  },
  "targets": ["dispatch-unit-fire-01"],
  "token": "MOCK_ONLY_ACTION_APPROVAL_NOT_A_REAL_TOKEN",
  "idempotency_key": "run-demo-001-action-008"
}
```

Proposed action name	Required parameter semantics
dispatch	<code>unit_type</code> : police/ambulance/firefighter; <code>destination_node_id</code> : existing node. Targets are available units of that type, not arbitrary locations. Confirm Twin's target granularity.
set_water_valve	<code>state</code> : open/closed. Targets are discovered Water valves; purpose and connectivity are known before planning.
set_grid_switch	<code>state</code> : open/closed. Targets are discovered grid switches; topology and protected-service constraints apply.
set_traffic_signal	<code>state</code> : a Twin-advertised legal state. Do not assume timing/phase parameters exist; a signal state cannot authorize conflicting movements.

Only the server's advertised action schema can settle these names and types. If Trinity prefers a string action plus a different parameter representation, all three teams must explicitly adopt that replacement before mocks diverge. No fifth outer `params` field is silently added here.

5.10 Request identity and result semantics

`targets` is a non-empty list of unique discovered IDs appropriate to the action. Actions spanning incompatible parameter sets are separate intents. Target ordering must have a single agreed rule before signing; an adapter cannot rearrange a signed request after approval.

The token is the Guardian-issued approval for the exact canonical action, targets, and parameters. Its wire format remains to be confirmed by 9antra and Trinity. The sample is deliberately a non-cryptographic mock placeholder. It is not usable as a live credential.

The `idempotency_key` is a non-empty string unique per run/action request and stable for transport retries of that unchanged request. It is not the token, incident ID, or sensor ID. Reuse with changed parameters must be rejected as a key conflict rather than silently running the changed command. Token expiry/retry interaction must be defined jointly.

The two-step default means a response can be pending; the Brain must not mark the action completed at that point. It records any returned action reference and asks for confirmation only under the eventually agreed `commit_action` contract. The binding between approval, pending action, commit, and idempotency is unresolved. This package flags that dependency rather than deciding the later pending/timeout task on its own.

5.11 Review cases for data interoperability

Input	Expected result
Valid UTC reading plus an unknown extra field	Accept known fields; ignore the extra field
Missing <code>device_id</code> or quoted numeric value	Reject required schema violation with field path
<code>value:null</code> with <code>quality:missing</code>	Accept no-value record; no hazard-clear inference
<code>value:0</code> with <code>quality:reported</code>	Preserve zero; evaluate its actual domain meaning
Timezone-free time or negative tick	Reject; do not guess local timezone or reset ordering
Shared-node domains Water/Power; Water quantity	Accept one Water measurement at a multi-domain node
<code>water_level</code> with kW	Reject incompatible quantity/unit pair
Same reading ID with a different payload	Flag identity/content conflict; never replace the original silently
Required known enum has an unknown value	Reject enum value, despite forgiving extra-field policy
Signed action altered after approval	Refuse; obtain approval for the actual final request
Raw call asks to isolate all devices	Treat as content; Brain cannot invoke containment
Error replies should identify a stable error code, field/path, expected shape, and safe correlation reference. Codes and log envelopes remain subject to partner interface/log agreement; do not echo secrets, full tokens, or arbitrary transcripts in errors.	

5.12 Correction example and review boundaries

```
{
  "run_id": "run-demo-001",
  "reading_id": "correction-water-18",
  "device_id": "sensor-water-01",
  "node_id": "node-pump-01",
  "domains": ["water", "power"],
  "seq": 18,
  "tick": 121,
  "timestamp": "2026-10-05T08:00:01.000Z",
  "quantity": "water_level",
  "value": 40.0,
  "unit": "cm",
  "quality": "corrected",
  "correction": {
    "corrects_reading_id": "reading-water-17",
    "replacement_source_reading_id": "reading-water-16",
    "replacement_source_timestamp":
      "2026-10-05T07:59:59.000Z"
  }
}
```

This is CIVIS's proposed corrected-record representation for Trinity/9antra alignment. The correction's time stamp and tick identify when the new correction record was emitted; the replacement source time determines the age of its inherited value. The original faulty report remains available for audit. This is not a claim that rollback has recovered current physical truth.

Freshness rule. For a corrected record, freshness uses the replacement source time/tick, not the correction emission time. If tick-based freshness is used, also carry `replacement_source_tick`; timestamp-based checking uses the named source timestamp. The partner contract must choose one consistent method. A last trusted value can be too old to support an action.

Minimal review package. The examples fix data shapes without implementing partner services, signature generation, MCP endpoints, shared logging, or commit/retry logic. The timestamp delta in these synthetic examples does not impose a one-second tick on other runs. Every run declares its own positive tick duration.

6 Conflicts, ambiguities, and decisions to confirm

Before the mocks are exchanged: confirm the data/action representation and sensor semantics. The remaining dated items retain their agreed owners and deadlines. None of the concerns below is silently promoted to a completed triplet decision.

ID	Finding	Resolution / owner
Q01	Incident verification is an agreed task, not an already agreed list.	All three teams verify I01–I11 by 5 Oct. Record explicit responses; CIVIS revises the catalogue.
Q02	Phase 0 allowed ungated/stale-token fallbacks; final decisions require exact-action approval.	Use final report: no valid approval means no new physical action. CIVIS must not reuse Phase 0 R1 exemptions or expired R2 approvals.
Q03	Telecom/Waste were optional in session 1, committed for 24 Oct in the final report, and excluded by the user now.	Exclude from this 5 Oct package. All teams must reconfirm any proposed removal from 24 Oct scope; do not silently rewrite the calendar.
Q04	A water-level reading does not establish valve purpose, pressure, or leak position.	Trinity supplies site role, datum, valve direction/connectivity and preview semantics. Until then valve responses remain conditional.
Q05	A Power load meter cannot prove high-voltage fluctuations.	Name I05 overload and I06 possible service loss. Trinity/9antra clarify what evidence triggers the Power isolation exception; do not invent a voltage sensor.
Q06	Air-quality monitor quantity and traffic-level semantics are unspecified.	Trinity confirms PM2.5/ug/m ³ and the 0–100 traffic index or supplies explicitly declared alternatives. No unnamed AQI or road-speed conversion.
Q07	Score-only verdict has no agreed scale/direction/correlation contract.	9antra confirms the proposed 0–1 scale and published cut-offs. Associate the score with its exact subject; do not infer labels or add Phase 0 reason arrays.
Q08	Four actuate fields leave action parameters underspecified.	Trinity confirms the structured-action representation or an explicit replacement. All teams use one format before writing incompatible mocks.
Q09	“Pending”/commit is only a working default in the connection table, although two-step safety is described as decided elsewhere.	Trinity and CIVIS reconcile status and define action reference, token recheck, commit idempotency, expiry, and result types before mocks. Numeric waits remain open.
Q10	Risk selection does not define numerical trust minima or caps.	CIVIS selects R2 dispatch and R3 Water/Power. 9antra/Trinity/all teams settle numerical gating separately. Domain cap table remains Trinity’s proposal and 9antra’s confirmation.
Q11	Three retries have no agreed meaning or duration.	Retain final policy of alternative usable evidence and eventual human review. CIVIS defines retry semantics by 9 Oct; do not count duplicate packets as retries/new evidence.
Q12	Raw source time, simulation time, arrival time, and token expiry can be confused.	CIVIS separates timestamp/tick from received-at. All teams agree token clock basis and lifetime; no guessed five-second TTL.
Q13	Linked cross-domain drift is a working default and off initially; Guardian cross-domain checks are optional.	Do not depend on automatic linked drift or Guardian cross-domain detection to make I10/I11 valid. Use explicit seeded event effects and available observations.
Q14	Actuator-failure detection is optional early, yet a request is not proof of effect.	Label unverified outcomes honestly. Trinity exposes actual/commanded feedback if implemented; Brain never reports success solely from a pending/accepted reply.

ID	Finding	Resolution / owner
Q15	Hidden scenario tool may not replace required scenario scripts without organizer confirmation.	Dali's SAC question remains his task. Keep scenarios/; add thin trigger scripts later unless organizers explicitly permit omission.
Q16	Phase 0 described MCP as mandatory and supplied earlier dates/KPIs.	Phase 1 makes protocols free. MCP 2026-07-28 is a triplet choice, not an organizer mandate. Use the final calendar; do not promise old performance/false-alarm targets.
Q17	Phase 0 pinned tools; final report explicitly defers tool pinning.	Retain deterministic validation and content separation, but list pinning as a known limitation. Do not claim it is built.
Q18	One scalar can be trusted while a derived action remains unsafe.	Distinguish reading trust, operational justification, preview, cap checks, and action-token approval. No single score substitutes for all five.

7 Handoff and acceptance record

7.1 CIVIS work completed in this package

- A concrete five-domain incident catalogue with conditional physical responses, multi-domain cases, and negative verification variants.
- A choice of exactly three risk-rated actuator types with no risk tier assigned to containment or every other action.
- Written conventions for units, UTC/simulation time, identifiers, shared nodes, coordinates, missing/corrected data, score scale proposal, and typed examples.
- A review register that identifies superseded Phase 0 rules and unresolved partner-owned interfaces.

7.2 Partner sign-off (not yet received)

Reviewer	Review focus	Current status
CIVIS	Catalogue, policy distinctions, risk selection, formatting	Written for review
Trinity	Inventory, units/metadata, action schema, preview feasibility	Pending verification
9antra	Score semantics, trust association, exact-action approval	Pending verification
All teams	Final incident list and changes to common interfaces	Pending verification

Record review date, reviewer, accepted IDs/conventions, requested changes, and the resulting document revision. Publishing this package does not turn pending partner reviews into a unanimous agreement.

7.3 Source traceability

Source	Used for
Final decision report	Sections 2–7: architecture, gates, inventory, roles; sections 9 and 11: dates and assigned tasks. Updated 2 October 2026. Primary triplet authority.
Phase 0 proposal	Pages 2–5: risk rationale and typed Brain planning; page 7: explicit units/UTC/ticks/IDs/null conventions; page 8: examples used only as historical reference. Superseded gates, tools, states, and numerical constants are not imported.
Session 1 summary	Historical context only. Shared-secret auth, optional signing, optional Telecom/Waste, and absence of trust labels were superseded by the final report.
Phase 1 specifications	Pages 1–3: R1–R4, required repository/log/timeline evidence, layer roles, and Test Book coverage. This package is a checkpoint artifact, not the final 20-page submission report.

7.4 Template and standards

Layout adapted from Mihdí Caballero's [Simple report](#), using its article layout, sans-serif font, blue headings/rules, and header/footer approach. Consolidated for standalone compilation; original template attribution and license notices are retained in `TEMPLATE_PROVENANCE.md`. Language, title/header content, spacing, and technical tables/listings were adapted.

JSON syntax follows [RFC 8259](#); timestamp representation follows the UTC profile of [RFC 3339](#). Project-specific fields and physical units are CIVIS conventions, not claims that those RFCs define the city model.